

Chun Huang

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RESEARCH PROFILE

I am a Ph.D. candidate in the Department of Physics at Washington University in St. Louis, advised by Prof. Alex Chen. My research centers on **neutron-star equation-of-state inference**, **X-ray pulse-profile modelling**, and **pulsar magnetosphere modeling**, connecting NICER/XMM-Newton observations, multimessenger constraints, and physics-based emission models.

Research snapshot

- **10 first-author papers**, including **3 single-author papers**.
- **6 corresponding-author / mentored student-led papers**, reflecting cross-institution supervision.
- **Software and infrastructure leadership**: lead developer and coordinator of `CompactObject`; active contributor to GPU-accelerated X-ray pulse-profile-modelling code `GPU-PPM`.

EDUCATION

- **Washington University in St. Louis**, St. Louis, MO
Ph.D. Candidate in Physics, Sep 2022 – May 2027 (expected)
Advisor: Prof. Alex Chen
- **Central China Normal University**, Wuhan, China
B.S. in Physics, Sep 2018 – Jun 2022

PUBLICATIONS (* denotes mentored student)

First-author papers

1. **Chun Huang**, “First-Principles Polar-Cap Currents in Multipolar Pulsar Magnetospheres,” *The Astrophysical Journal* **999**, 204 (2026). DOI: 10.3847/1538-4357/ae4489
2. **Chun Huang**, “Model-Independent Determination of the Tidal Deformability of a $1.4 M_{\odot}$ Neutron Star from Gravitational-Wave Measurements,” *The Astrophysical Journal* **985**(2), 2025. DOI: 10.3847/1538-4357/add5ef
3. **Chun Huang** and Alexander Chen, “Physics-Motivated Models of Pulsar X-ray Hotspots: Off-center Dipole Configurations,” *The Astrophysical Journal* **991**, 90 (2025). DOI: 10.3847/1538-4357/adf747
4. **Chun Huang** and Shashwat Sourav*, “Constraining First-order Phase Transition inside Neutron Stars with Application of Bayesian Techniques on PSR J0437–4715 NICER Data,” *The Astrophysical Journal* **983**(1), 2025. DOI: 10.3847/1538-4357/adbb67
5. **Chun Huang**, “Equation-of-State-Independent Determination of the Radius of a $1.4 M_{\odot}$ Neutron Star Using Mass-Radius Measurements,” *The Astrophysical Journal Letters* **978**(1), 2025. DOI: 10.3847/2041-8213/ad9f3c
6. **Chun Huang**, Tuhin Malik, João Cartaxo, Shashwat Sourav, Wenli Yuan, Tianzhe Zhou, Xuezhi Liu, John Groger, Xieyuan Dong, Nicole Osborn, Nathan Whitsett, Zhiheng Wang, Constança Providência, Micaela Oertel, Alexander Y. Chen, Laura Tolos, and Anna Watts, “CompactObject: An Open-source Python Package for Full-scope Neutron Star Equation-of-State Inference,” submitted to *Journal of Open Source Software* (2024). arXiv:2411.14615
7. **Chun Huang**, Laura Tolos, Constança Providência, and Anna Watts, “Constraining a Relativistic Mean Field Model Using Neutron Star Mass-Radius Measurements II: Hyperonic Models,” *Monthly Notices of the Royal Astronomical Society* (2024). DOI: 10.1093/mnras/stae2792

8. **Chun Huang** and Xiaoping Zheng, “Bayesian Insights into Post-glitch Dynamics: Model Comparison and Parameter Constraint from Decades-long Observation Data of the Crab Pulsar,” *Monthly Notices of the Royal Astronomical Society* **542**, 3198–3205 (2025). DOI: 10.1093/mnras/staf1415
9. **Chun Huang**, Geert Raaijmakers, Anna Watts, Laura Tolos, and Constança Providência, “Constraining a Relativistic Mean Field Model Using Neutron Star Mass-Radius Measurements I: Nucleonic Models,” *Monthly Notices of the Royal Astronomical Society* **529**(4), 4650–4665 (2024). DOI: 10.1093/mnras/stae844
10. **Chun Huang** and Yuchen Jiang*, “Symmetry Breaking Paradigm in Typical Laminar–Turbulence Transition System,” preprint (2021). arXiv:2108.09486

Corresponding-author / mentored student-led papers

11. Tianzhe Zhou* and **Chun Huang**, “GPU-Accelerated X-ray Pulse Profile Modeling,” *A&A* **709**, A111 (2026). DOI: <https://doi.org/10.1051/0004-6361/202659305>
12. Tianzhe Zhou* and **Chun Huang**, “Hidden Twin-star Solutions from an Agnostic Speed-of-sound Model: XTE J1814–338 and GW170817 as a Multimessenger Differentiator,” *The Astrophysical Journal* **1002**, 120 (2025). DOI: 10.3847/1538-4357/ae552e
13. Ryan O’Connor*, **Chun Huang**, and Alexander Chen, “Combining the Mass–Radius Posteriors of J0030+0451 Allowing for Unknown Model Systematics,” submitted to *The Astrophysical Journal* (2026). arXiv:2604.11621
14. Xue-Zhi Liu*, Premachand Mahapatra, **Chun Huang**, Ayush Hazarika, Chiranjeeb Singha, and Prasanta Kumar Das, “Revealing Dark Matter’s Role in Neutron Stars Anisotropy: A Bayesian Approach Using Multi-messenger Observations,” *Physical Review D* **112**, 083032 (2025). DOI: <https://doi.org/10.1103/zhs6-487x>
15. João Cartaxo*, **Chun Huang**, Tuhin Malik, Shashwat Sourav, Wenli Yuan, Tianzhe Zhou, Xuezhi Liu, and Constança Providência, “Covariant Energy Density Functionals for Modeling the Equation of State of Neutron Star Matter: Cross-comparison Analysis Using CompactObject,” *The Astrophysical Journal Supplement* **282**, 33 (2025). DOI: 10.3847/1538-4365/ae2310
16. Wen-Li Yuan*, **Chun Huang**, Chen Zhang, Enping Zhou, and Renxin Xu, “Bayesian Inference of Strangeon Matter Using the Measurements of PSR J0437–4715 and GW190814,” *Physical Review D* **111**, 063033 (2025). DOI: 10.1103/PhysRevD.111.063033

Selected collaborative papers

17. Xiaoping Zheng, Weihua Wang, **Chun Huang**, Jianping Yuan, and Sheng-Jie Yuan, “The Persistent Shift in Spin-down Rate Following the Largest Crab Pulsar Glitch Rules Out External Torque Variations due to Starquakes,” *The Astrophysical Journal* **982**(2), 2025. DOI: 10.3847/1538-4357/adbca6
18. Kangkang Shao, **Chun Huang**, and Qin Qin, “Data Determination of HQET Parameters in Inclusive Charm Decays,” *The European Physical Journal C* **85**, 1011 (2025). DOI: 10.1140/epjc/s10052-025-14691-z

RESEARCH LEADERSHIP, FUNDING, AND MISSION SCIENCE

- **Lead developer and coordinator, CompactObject collaboration** — coordinated an international collaboration of researchers across the U.S., Europe, and China to build an open-source Bayesian inference framework for neutron-star equation-of-state and compact-object studies.
- **Proposal lead writer and scientific lead, awarded NICER proposal** — “GPU-Accelerated Pulse-Profile Analysis of NICER MSPs with Physics-Based Hotspot Models” (2026–2027). Led the scientific concept and writing; PI listed as faculty advisor because of graduate-student PI restrictions.
- **Major scientific contributor and co-writer, NASA ADAP proposal** — “Constraining the Physical Properties of Magnetars using their Pulsed X-ray Emission” (2026 submission), with primary contributions in the soft X-ray / NICER modeling and Bayesian-inference program.

- **NewAthena WG4 mission-science special issue contributor** — abstract accepted and merged into the NewAthena JHEAP Special Issue paper under the provisional title “NewAthena constraints on the dense matter equation of state via X-ray pulse profile modelling of millisecond pulsars”; all merged-paper authors share **corresponding-author status**.

HONORS AND TRAVEL AWARDS

- Chambliss Student Poster competition finalist, AAS 248 Summer meeting (2026)
- Dean’s Award for Graduate Research Excellence, semi-finalist, Washington University in St. Louis \$500 (2025)
- Study Abroad Scholarship for Outstanding Students, China Scholarship Council (exceptional top 20 globally), \$10,000 (2025)
- Outstanding Undergraduate Thesis Award, Wuhan, China (2022)
- First Prize, “Lun Dao” Doctoral Student Forum, SJTU Tsung-Dao-Lee Institute (sole undergraduate awardee) (2021)
- Best Report Second Prize (ranked 8/40), PKU-DoA Undergraduate Astronomy Symposium (2021)

SELECTED TALKS AND PRESENTATIONS

- June 2026 — Poster Presentation, AAS 248 Summer meeting at Pasadena
- May 2026 — Selected Talk, CeNAM Frontiers in Nuclear Astrophysics Meeting 2026
- May 2026 — Poster Presentation, Purdue Plasma Physics Conference
- Oct 2025 — Oral Presentation, HEAD 22 meeting, St. Louis, MO
- Sep 2025 — Invited Talk, Harvard CfA seminar
- Sep 2025 — Invited Talk, ITC luncheon, Harvard ITC
- Sep 2025 — Invited Talk, MAT talk, MIT Kavli Institute
- Aug 2025 — Seminar Talk, AstroPlasma Seminar, Princeton Astronomy
- Aug 2025 — IAS Astro coffee talk, Institute for Advanced Study
- Aug 2025 — Invited Talk, Multi-RAPTOR forum, UC Berkeley
- Aug 2025 — Invited Talk, KIPAC Tea, Stanford Astronomy
- Jun 2025 — Invited Talk, UCLA Physics seminar (remote)
- May 2025 — Selected Talk, IReNA-CeNAM 2025: Frontiers in Nuclear Astrophysics Meeting
- May 2025 — Selected Talk, Princeton PCTS workshop, “Extreme Physics of Neutron Star Interiors”
- Mar 2025 — Invited Seminar, THEA Seminar, Columbia University Department of Astronomy
- Nov 2024 — Group Meeting Talk, Nicolas Yunes Group, UIUC
- Jun 2024 — Astrophysics Seminar, Peking University
- Jun 2024 — Seminar Talk, Tsinghua Astronomy
- May 2024 — Group Meeting Talk, University of Groningen (remote)
- Mar 2024 — Seminar Talk, Tufts University
- Mar 2024 — Invited Talk, MIT Kavli Institute
- Mar 2024 — Seminar Talk, Harvard CfA

Other contributed talks, posters, and outreach

- Feb 2025 — Invited interview, AAS Journal Author Series
- Jun 2024 — Poster Presentation, Purdue Plasma Physics Conference
- Jun 2023 — Contributed Talk, “Neutron Star in Heaven and Earth,” INT workshop, University of Washington
- Dec 2021 — Contributed Talk, “Lun Dao” Doctoral Student Forum, Tsung-Dao-Lee Institute
- Sep 2021 — Contributed Talk, PKU-DoA Undergraduate Astronomy Symposium
- Aug 2021 — Presentation, 5th Round Workshop on Theoretical Astrophysics, Lanzhou University
- Aug 2021 — Presentation, Huazhong University of Science and Technology
- Jun 2020 — Presentation, Xiaosong Chen’s Group, Beijing Normal University

MENTORING AND TEACHING

Research mentoring

- **Undergraduate and master’s students (co-advisor / mentor)**
 - Xuezhi Liu — M.S. (Central China Normal University); 1 *PRD* published
 - Ryan O’Connor — B.S. (Washington University in St. Louis; co-advised with Prof. Alex Chen)
- **Graduate students and postdocs (project lead; idea generation and mentorship)**
 - Wen-Li Yuan — Postdoc (Peking University); 1 *PRD* published, 1 *PRD* in press
 - Tianzhe Zhou — Ph.D. student (Tsinghua University); 1 *A&A* published, 1 *ApJ* published
 - Shashwat Sourav — Ph.D. student (Washington University in St. Louis); 1 *ApJ* published

Teaching

- Teaching Assistant, Washington University in St. Louis 2022–2025
Physics 191 and 192 laboratories (introductory physics)

SCIENTIFIC COLLABORATION AND PROFESSIONAL SERVICE

- MUSES (Modular Unified Solver of the Equation of State) Collaboration, since 2024
- Referee for *The Astrophysical Journal*, *Frontiers in Astronomy and Space Sciences*, *Classical and Quantum Gravity*, and *Physica Scripta*